The online platform for area-wide management of Western X-disease

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Introduction

Western X disease is a plant disease that affects stone fruit trees, like cherries, peaches, plums, and apricots. It is caused by a phytoplasma, a type of bacterium.

Infection: The disease is spread by leafhoppers that feed on the sap of infected plants and then move on to healthy plants.

Symptoms: Infected trees show signs like yellowing leaves, poor fruit quality, and stunted growth. The leaves might curl or become discolored, and the fruit might be smaller or misshapen.

Credits to treefruit.wsu.edu

Introduction

Impact: Over time, the disease can weaken the tree, reduce fruit production, and eventually kill the tree. Lost revenues valued at $65 million and generated re-establishment costs estimated at $115 million during 2015-2020 epidemic in Washington and Oregon states.

Management: There is no cure for Western X disease. Management focuses on preventing the spread by controlling leafhoppers and removing infected trees to protect healthy ones.

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Program Relevance

- Studies show that removing infected trees reduced the infection rate across orchards by 65%.

- However, growers tend to keep the infected trees longer to obtain extra cherry harvest at the cost of increased disease spread and damages in future seasons. Uncoordinated insecticide applications and delayed tree removal exacerbate the problem.

- Due to high mobility of X-disease vector, collaborative efforts of growers are highly recommended to coordinate their management actions across orchard boundaries to manage the disease and its vectors.

Target Audience

- Decision-makers in sweet cherry growing regions of Washington, California, Idaho, and Oregon.

- Owners, managers, commercial disease consultants, and extension officers.

This program provides tailored strategies for each group of growers based on their local characteristics.

Behavioral Change and Program Outcomes

Learning Objectives:

- Educate growers on using a web-based platform for AWM.

Application of Information:

- Growers will use platform data to make informed decisions on tree removal.
- Providing transparent information about the program
- Policy makers use data to monitor the impact of AWM program.

Behavioral Change and Program Outcomes

Specific Behavioral Change:

- Coordinated disease management through timely removal of infected trees.

Evaluation Plan:

- Surveys and feedback mechanisms to assess knowledge, platform usability, and program impact.

Analytical Research Foundation - Literature

1. Bioeconomic models of private disease control

Fenichel and Horan (2007); Horan and Wolf (2005); Gramig and Horan (2011); Atallah et al. (2015).

- Population is spatially imperfectly-mixed (different probability of infection for individuals in different positions in network)
- Individuals are heterogeneous in their disease ecologic parameters.

2. Bioeconomic models of cooperative and noncooperative disease control among neighboring farms

Bhat and Huffaker (2007); Fenichel, Richards, and Shanafelt (2014); Costello et al. (2017); Epanchin-Niell and Wilen (2015), Atallah et al. (2017); Siriwardena et al. 2018; Martinez et al. (2023)

- Strategic complementarity or strategic substitution and free riding in control

Analytical Research Foundation – Economic Model

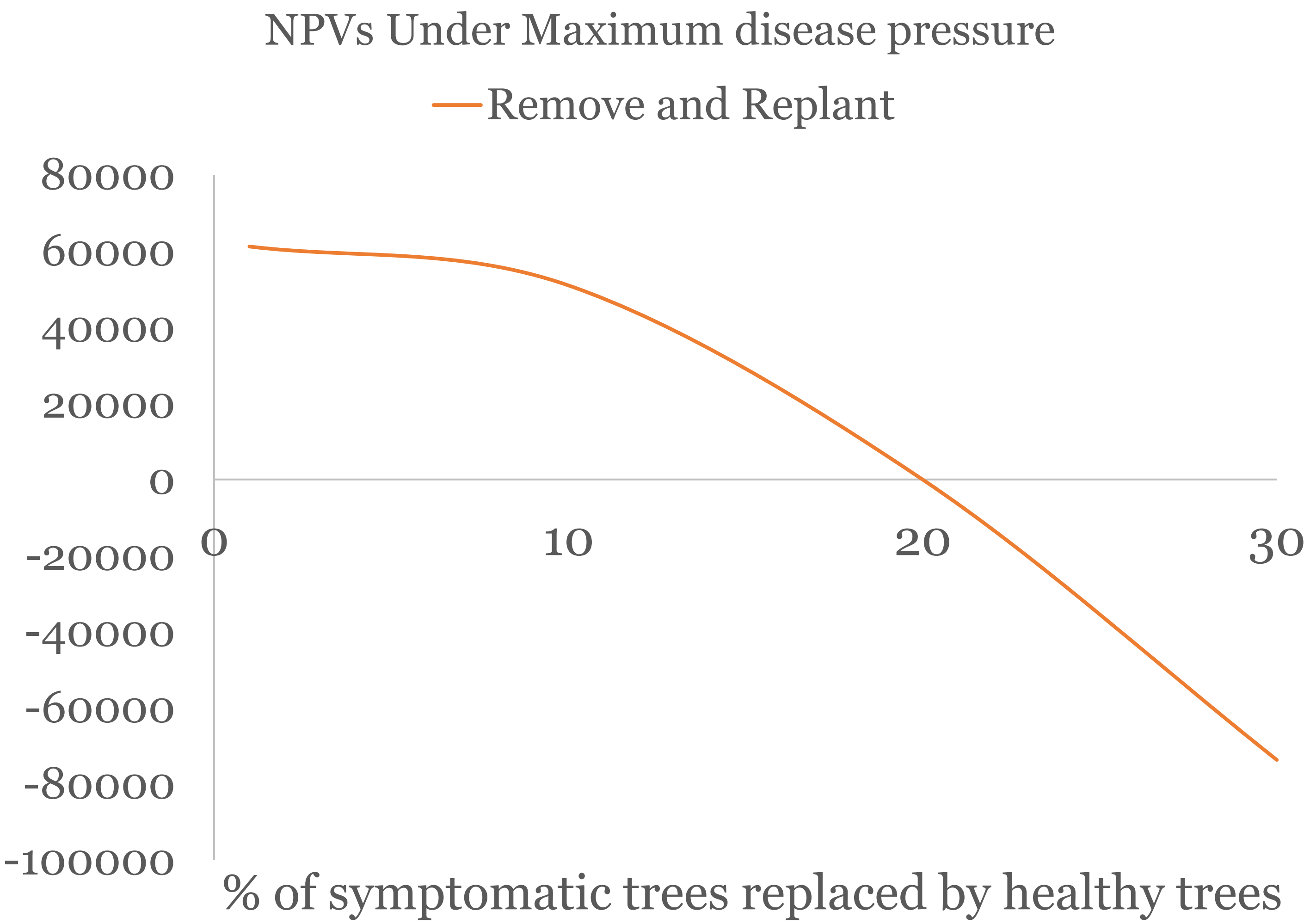
A farm-level bioeconomic model that integrates X-disease dynamics, crop growth, with a profit-maximization model of optimal tree removal and replanting decision

$$NPV = \sum_{t}^{T}\sum_{i}^{I}\sum_{j}^{J} py(u_{i,j,t})_{t,i,j} d\left(\underline{E(s_{t,i,j})}\right) - c(u_{i,j,t})_{t,i,j}$$

- (i, j) is the location of a tree on a grid with $i = 20$ rows and $j = 30$ columns representing an orchard with 600 trees.
- $\boldsymbol{u_{i,j,t}}$ **is a binary decision variable (removing and replanting)**.
- $d((E(s_{t,i,j})))$ is a damage function which maps expected health of tree to a corresponding yield damage factor.

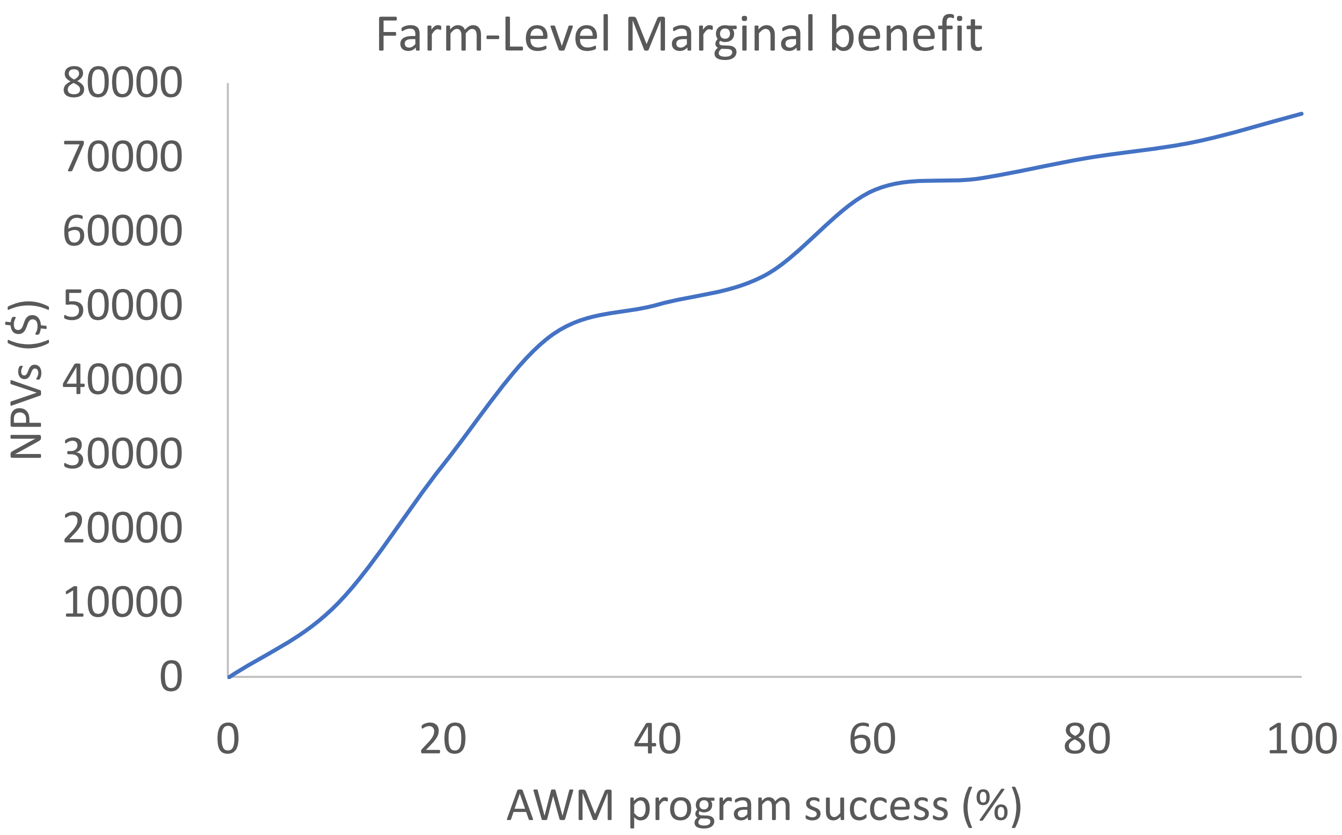
Interpretation of Results: removal and replanting

- Optimal action
- Can wait until 20%



Interpretation of Results: area-wide management program

- AWM decreases disease pressure
- AWM increases individuals' profits



Delivery Materials and Methods

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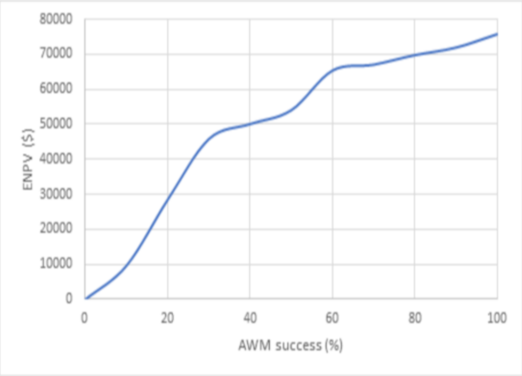
Orchard Analysis

Orchard-level action	Expected profit over the orchard's lifespan		Analysis
	Low disease pressure	High disease pressure	
Tree Removal Ratio (%)	-32,319.33	-209,478.67	You will lose money if you do not manage the infected trees.
10	94,131.70	51,001.08	Your economic returns are positive if 10% of your trees are symptomatically infected.
20	70,586.29	152.91*	*Your economic return is not different from zero if 20% of your trees are symptomatically infected. 20% is the break-even point for your orchard. Based on the current characteristics of your orchard, we recommend you remove and replant symptomatically infected trees at the first onset. You need to clear cut the orchard and replant new trees, once you have more than 20% symptomatically infected trees in your orchard.
30	21,353.51	-73,659.90	Your economic returns are negative if 30% of your trees are symptomatically infected and there is a high disease pressure in your area.

Area-wide management outlook



This graph shows the economic impact of AWM on your business. This scenario considers the case where participants in AWM program will allow at most 20% symptomatically infected trees in their orchard. The marginal benefit of AWM to your business is decreasing as more growers are committing to the program.

Delivery Materials and Methods

Webinars

Social Media

YouTube Videos

Presentations at growers' meetings

Extension Articles at popular journals and platforms

Delivery Materials and Methods

Innovation:

- Real-time data sharing and recommendations based on bioeconomic models
- Tailored to orchard and grower Individual characteristics

Delivery Challenges:

- Ensuring grower participation and commitment
- Low familiarity with online decision tools

Thank You!

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